A Hitchhiker's Guide to AI in MS

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MSToronto2026

10th Joint ACTRIMS-ECTRIMS Meeting

OCTOBER 21-23, 2026 | TORONTO, CANADA

iAIMS

international Community
for AI in Multiple Sclerosis

Origin of AI

The “Imitation Game”



Alan Turing

1950

2026

https://nl.wikipedia.org/wiki/Alan_Turing

Dartmouth workshop



Basics of “Artificial Intelligence” established, term first coined



1950

1956

2026

Ray Kurzweil – “Architects of Intelligence”
https://spectrum.ieee.org/dartmouth-ai-workshop

Dartmouth workshop



Basics of “Artificial Intelligence” established, term first coined

Marvin Minsky

“The field of research concerned with making machines do things that people consider to require intelligence.”

Claude Shannon

First name rings a bell?



1950 1956 2026

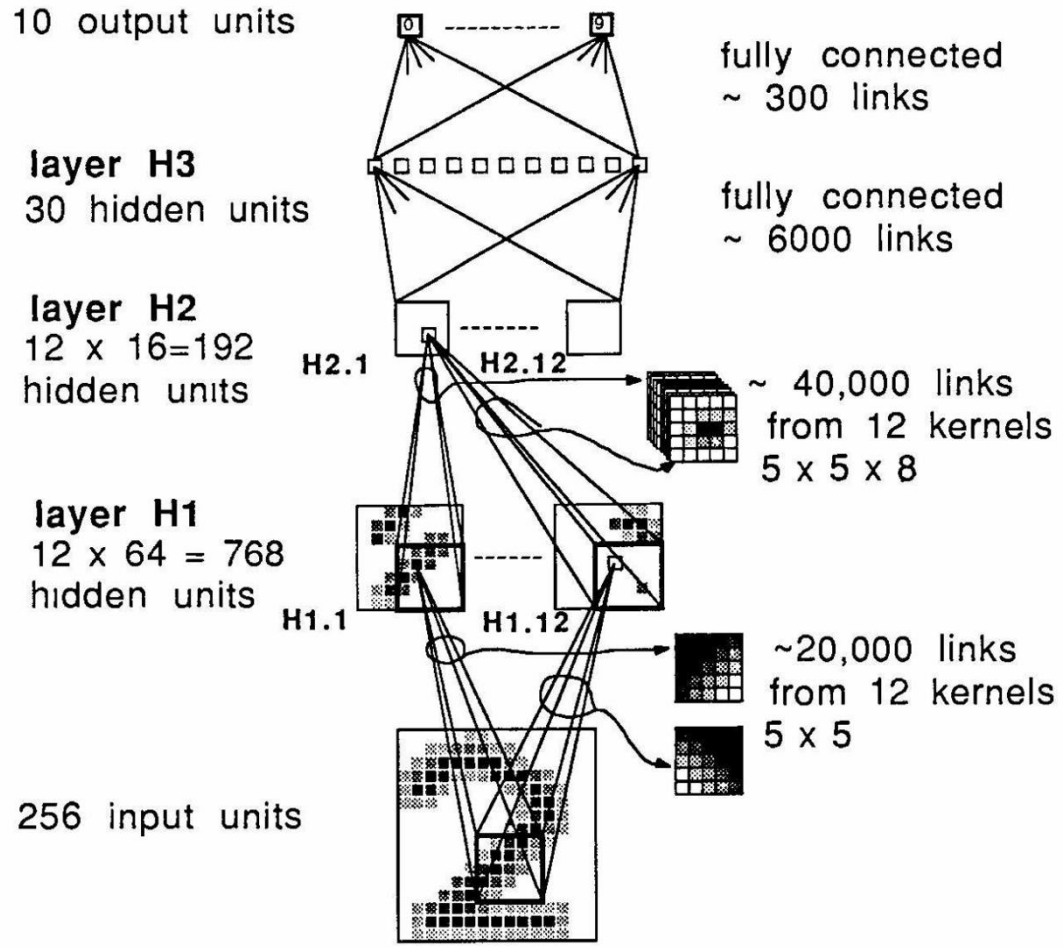
Ray Kurzweil – “Architects of Intelligence”
https://spectrum.ieee.org/dartmouth-ai-workshop

The first CNN



Yann Lecun

Convolutional Neural Network (CNN)



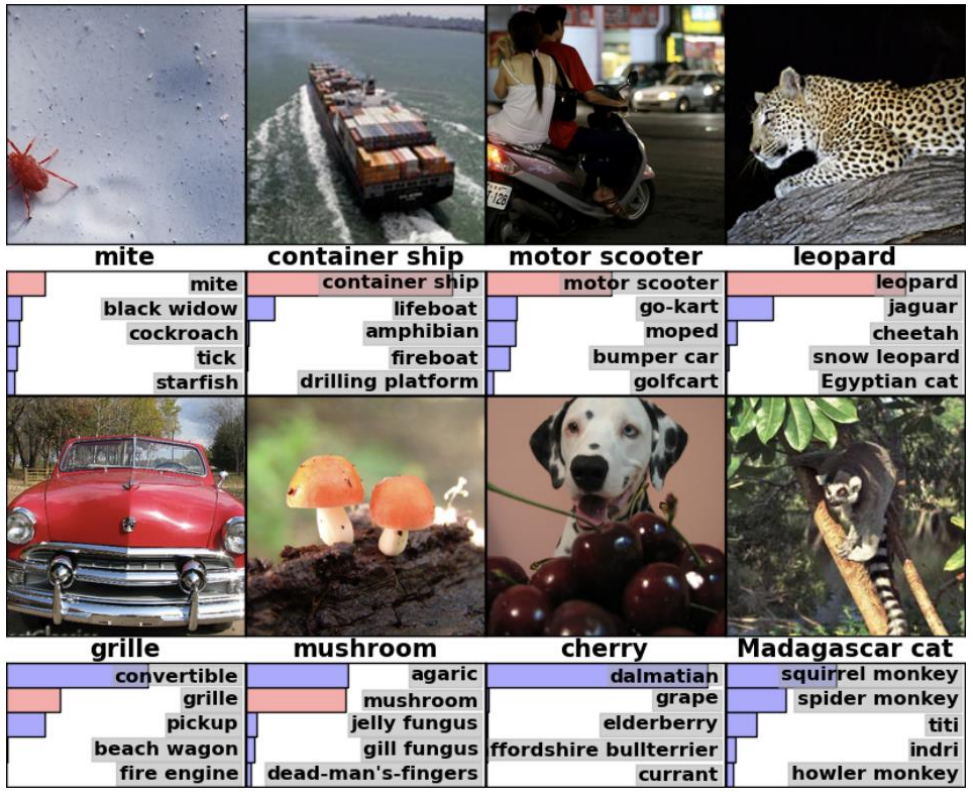




1950 1956 1989 2026

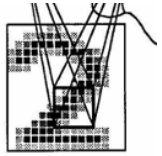
LeCun et al. 1989

Waiting for power ...









1950 1956 1989 2012 2026

Krizhevsky et al. 2012

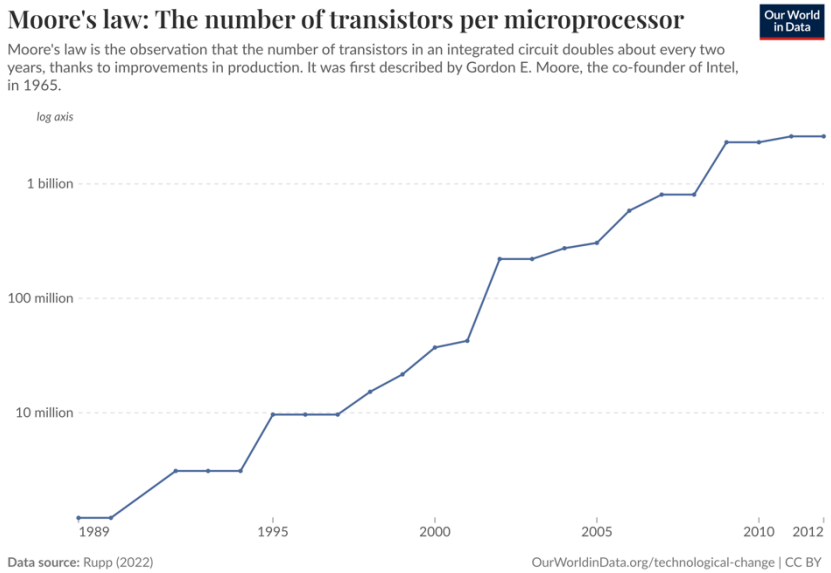
Waiting for power ...

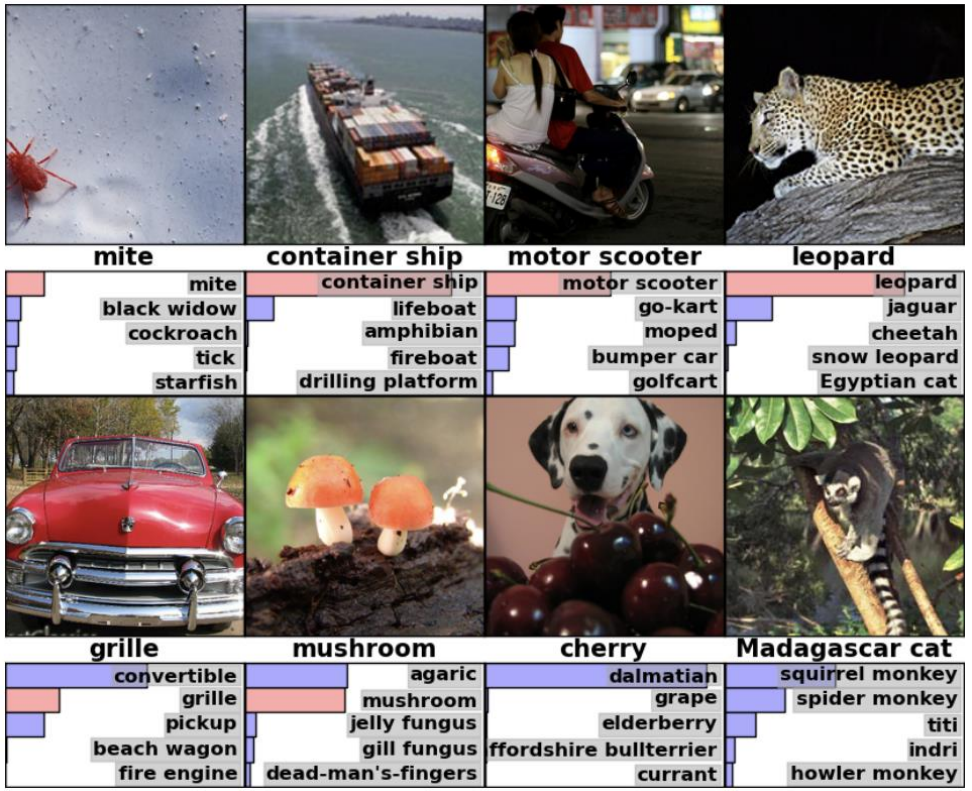
1989: 1.21 million

x 2149

2012: 2.60 billion

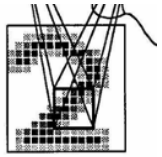
Graphical Processing Units (GPUs)











1950 1956 1989 **2012** 2026

Krizhevsky et al. 2012

https://ourworldindata.org/grapher/transistors-per-microprocessor

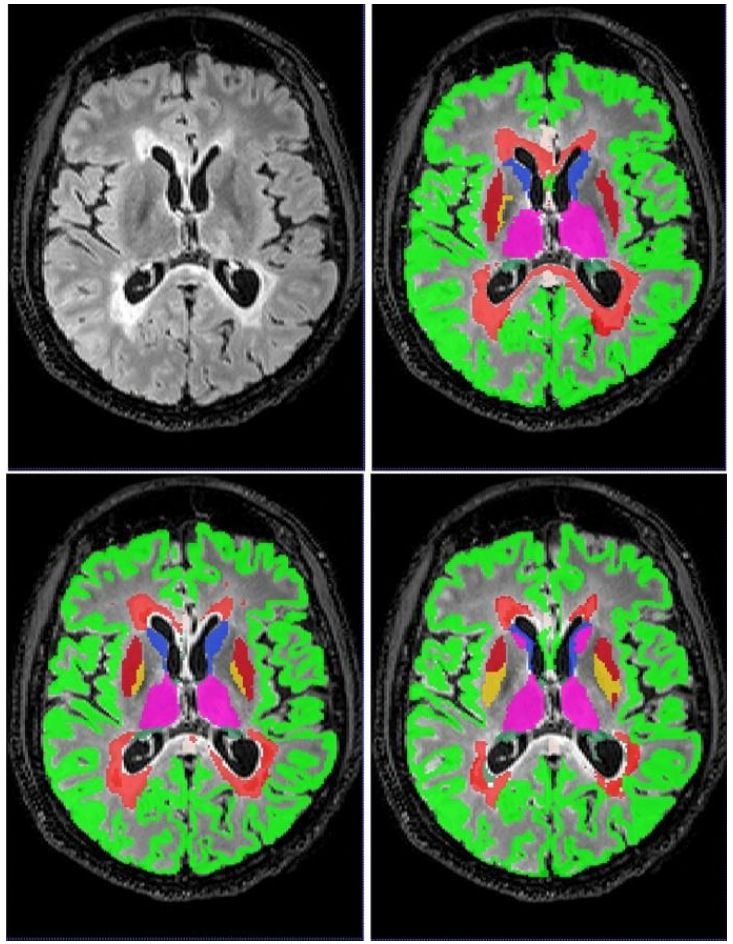
MS interlude: AI in MS much about imaging

14 FDA-approved Medical Devices

- All radiological tools
- Improved image analysis: 12
- Portable MRI device: 1
- Image enhancement software: 1



More on imaging:
Prof. Bach Cuadra



McKinley et al. 2021

When things changed for text

Attention Is All You Need

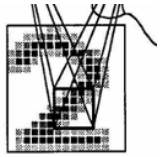
"Transformer" – The "T" in ChatGPT



Ashish Vaswani







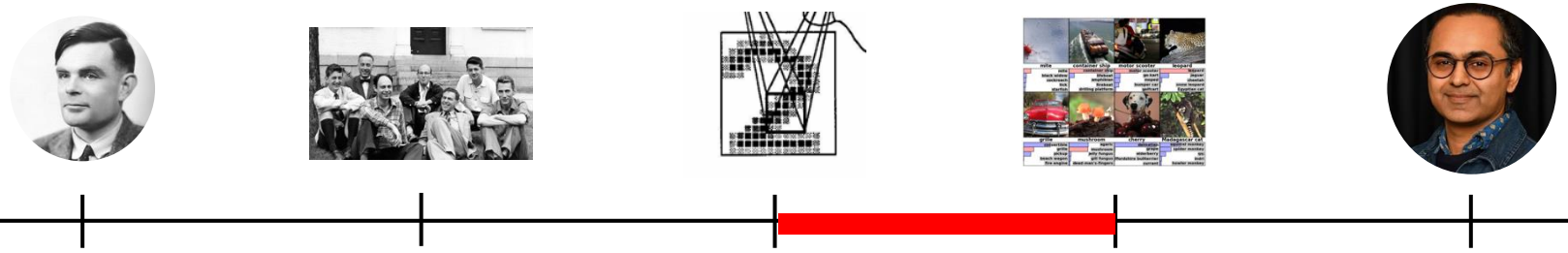


1950 1956 1989 2012 2017 2026

Vaswani et al. 2017 - NIPS

When things changed for text



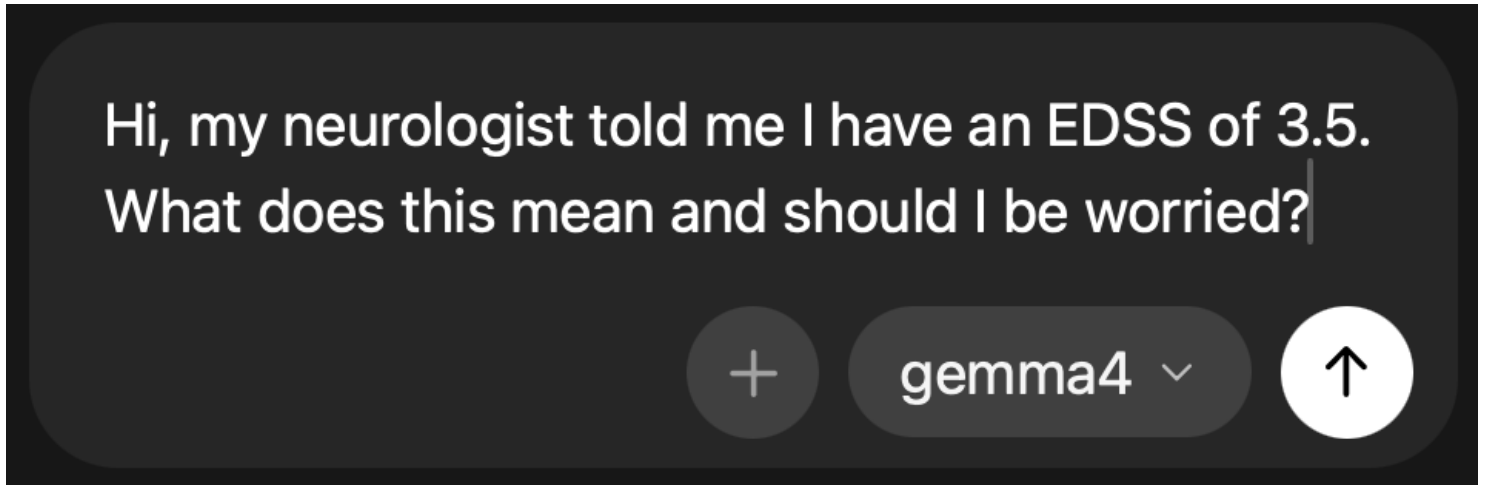


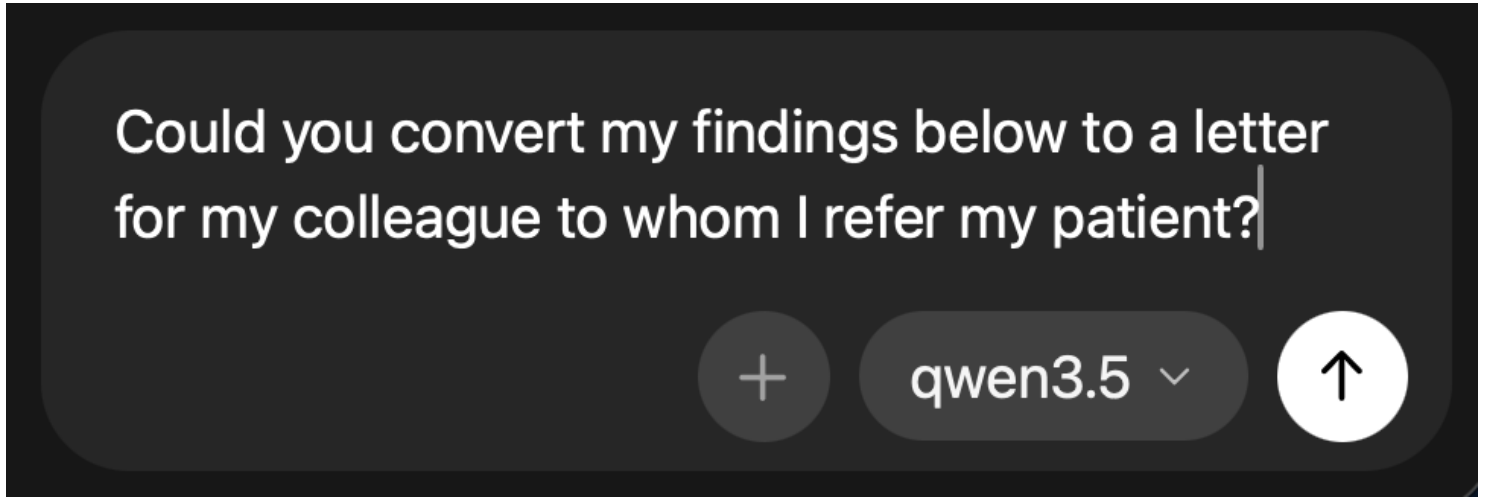
https://www.threads.com/@whosthat360/post/DH2ujK5y--a/sam-altman-took-to-x-formerly-twitter-saying-can-yall-please-chill-our-team-need

MS interlude: Chatbots enter doctor's room

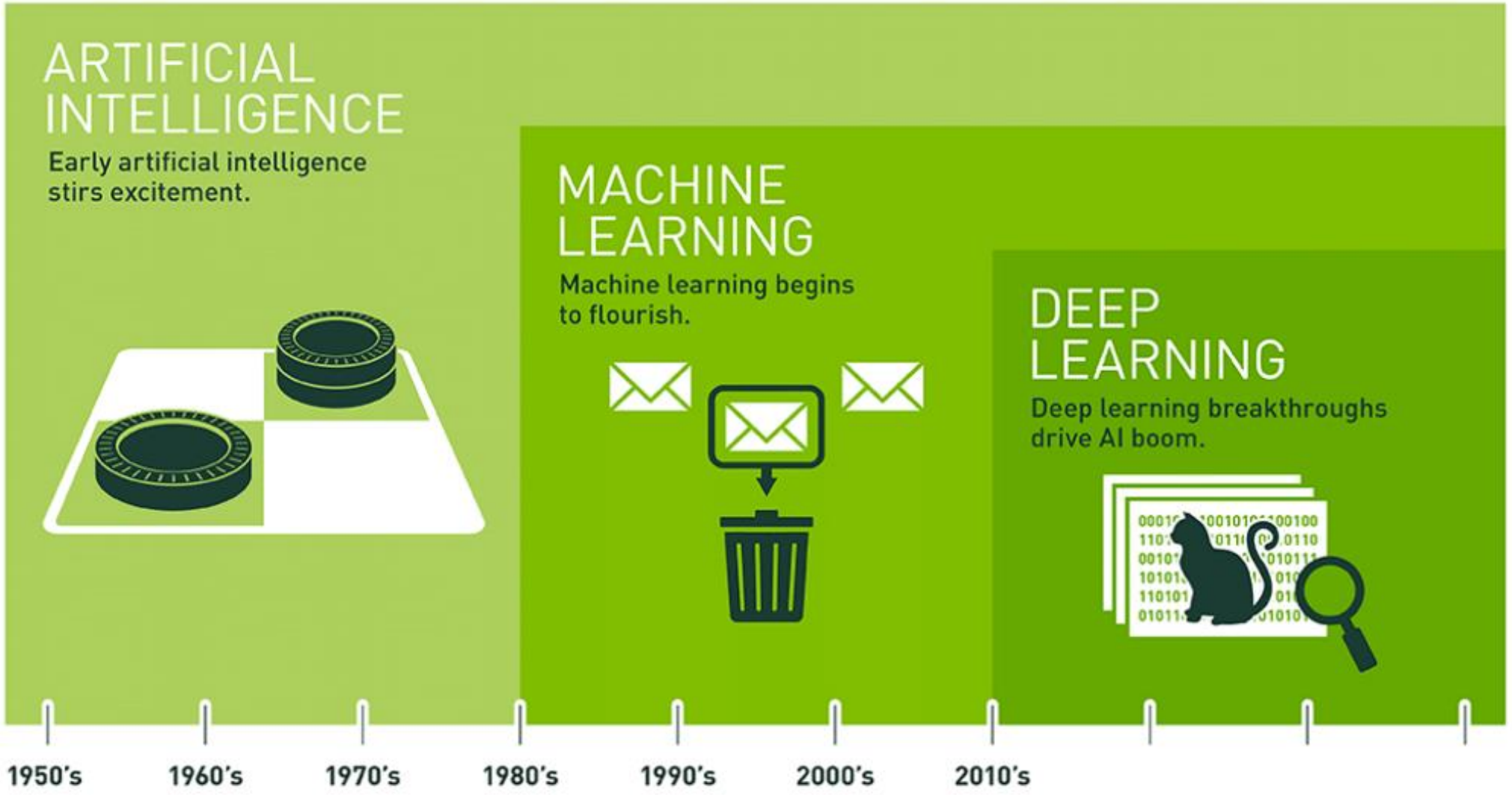


More on Large Language Models (LLMs):
Dr. Broeders





AI concepts in brief





Tabular



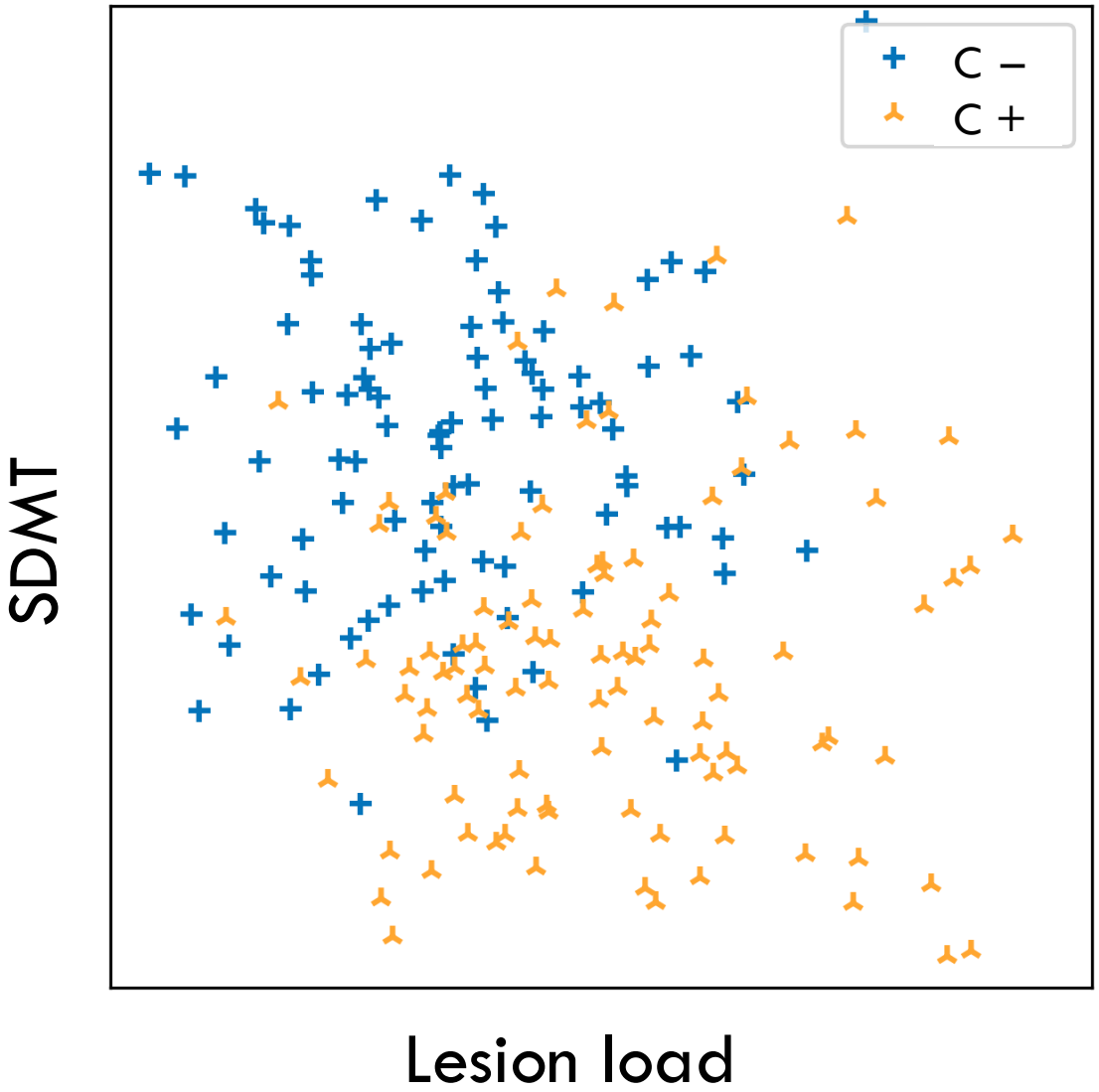
Imaging



Text

Machine learning: trying to learn function y = f(x) from data

A clinically relevant question in MS



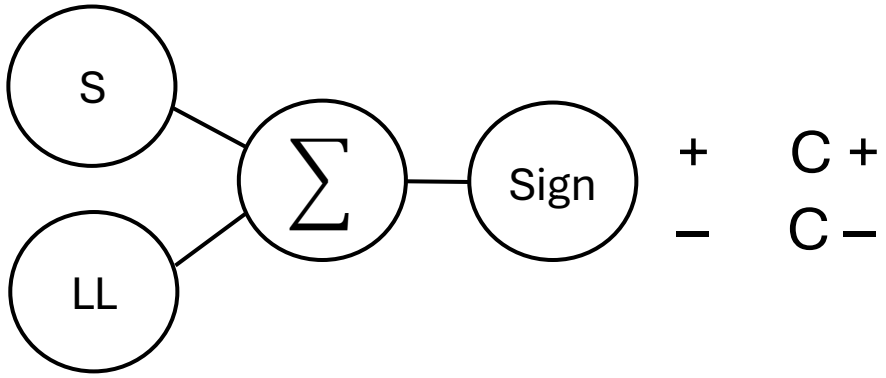
Goal: Predict conversion

C – RRMS > RRMS

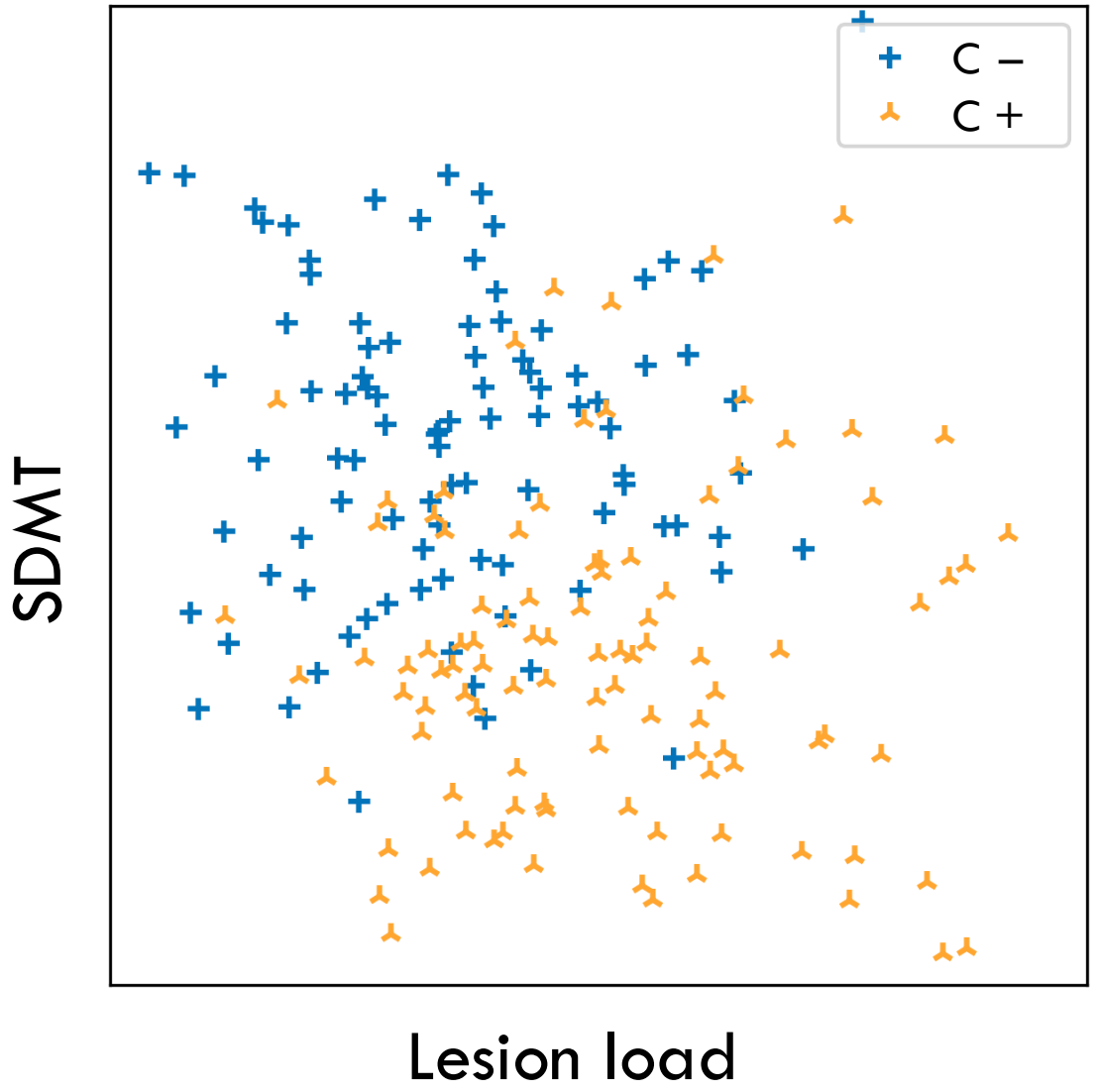
C + RRMS > SPMS

Welcome to iAIMS's practice

Conversion = sign(S*$\mathbf{w_S}$ + LL*$\mathbf{w_{LL}}$)

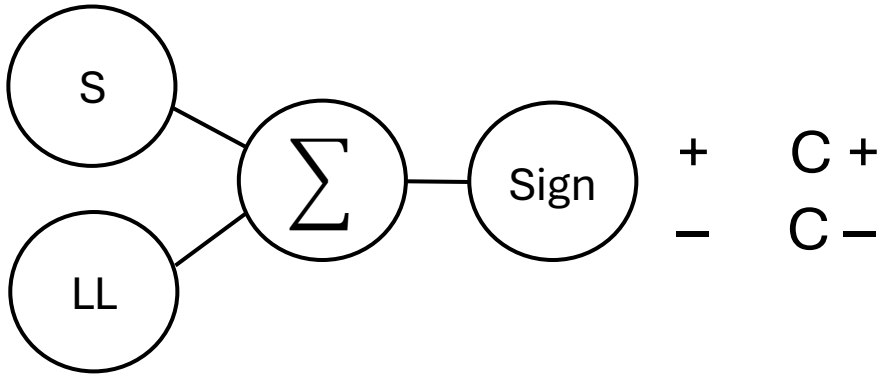


Training data



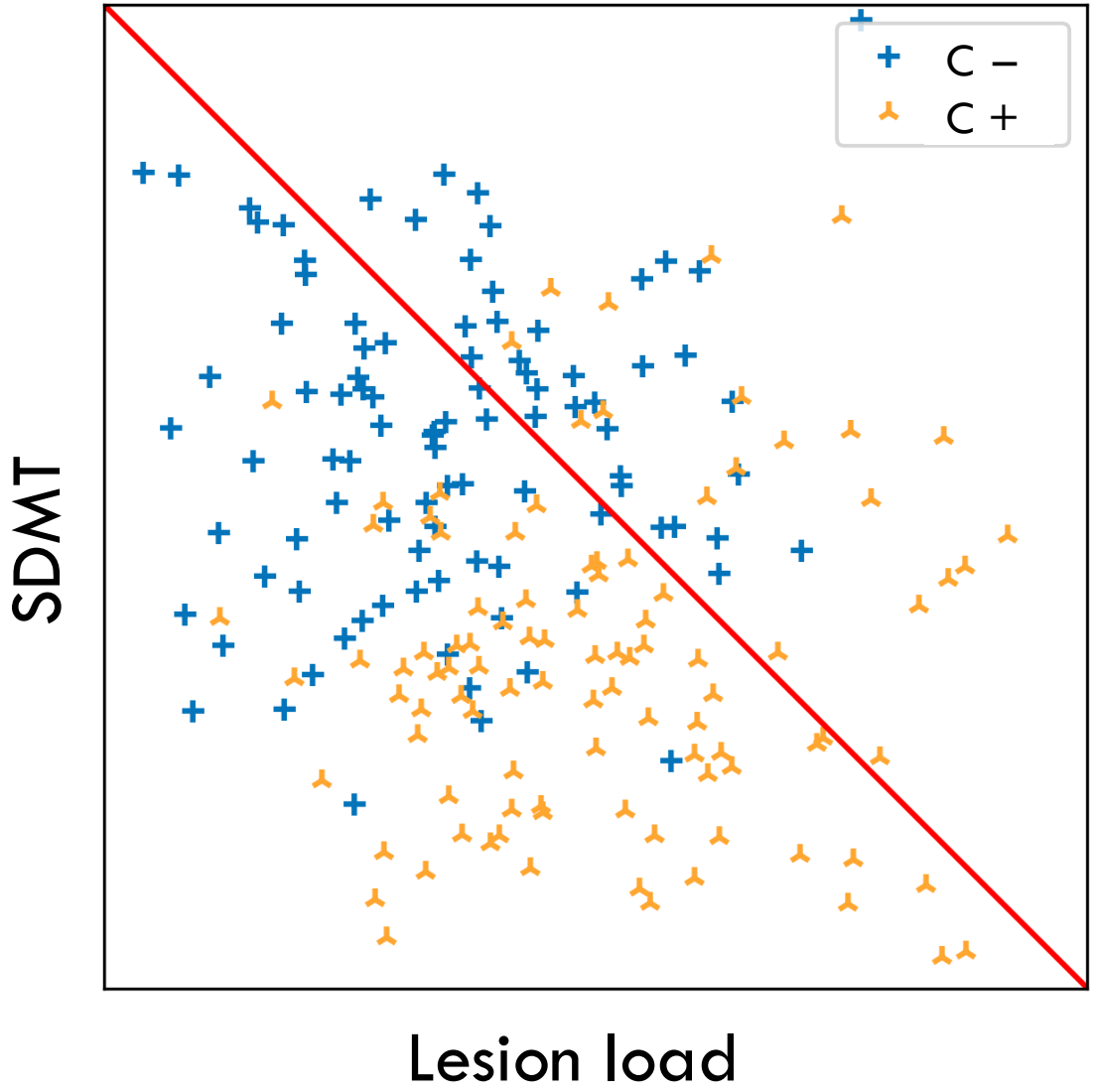
Our first guess

Conversion = sign(S*w_S + LL*w_{LL})



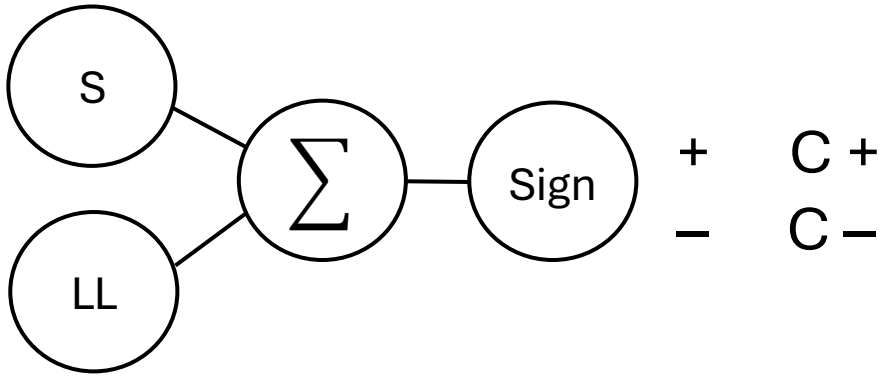
$w_S = 1$

$w_{LL} = 1$



Our first guess

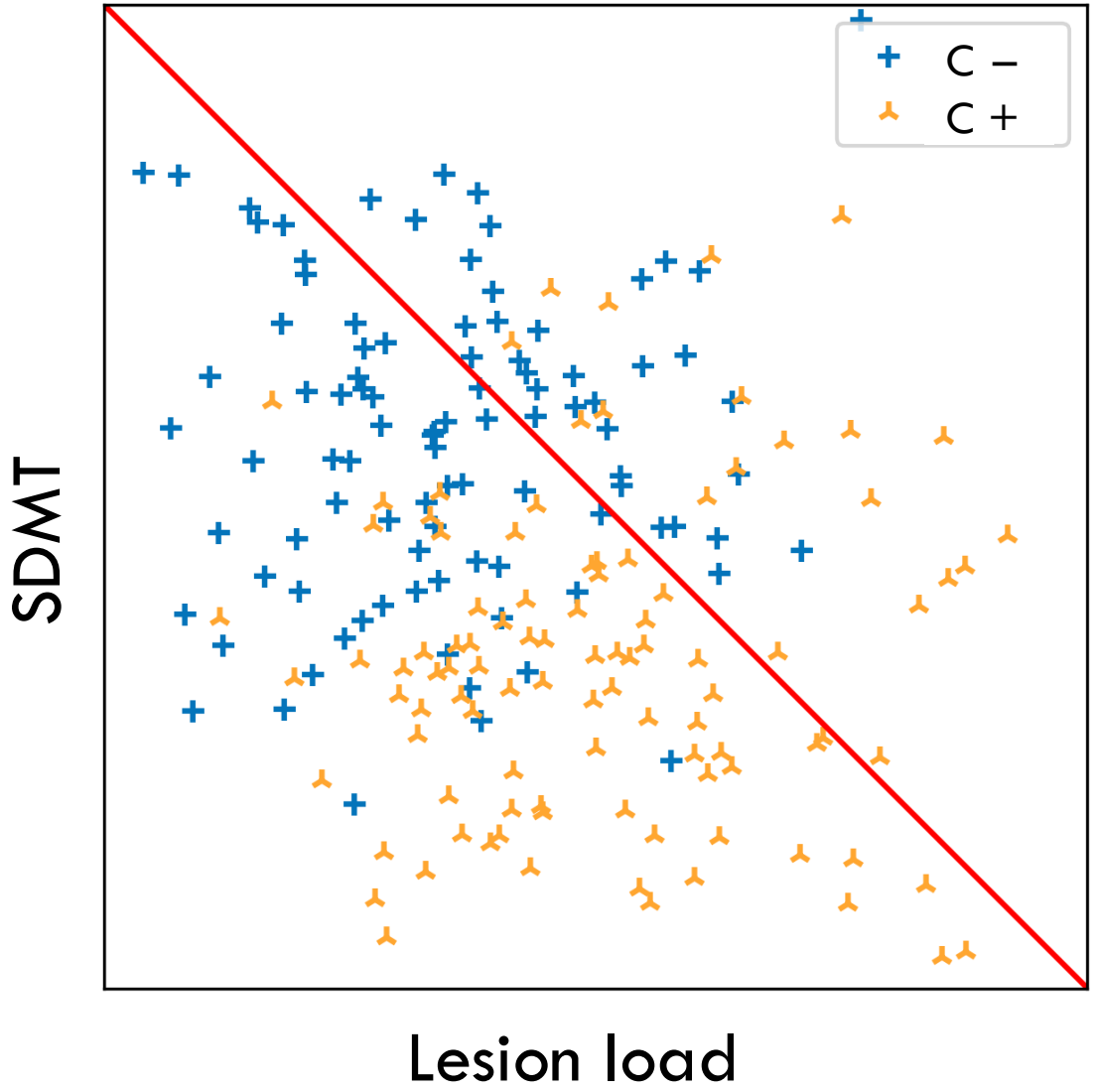
Conversion = sign(S*w_S + LL*w_{LL})



$w_S = 1$

$w_{LL} = 1$

Pretty terrible…



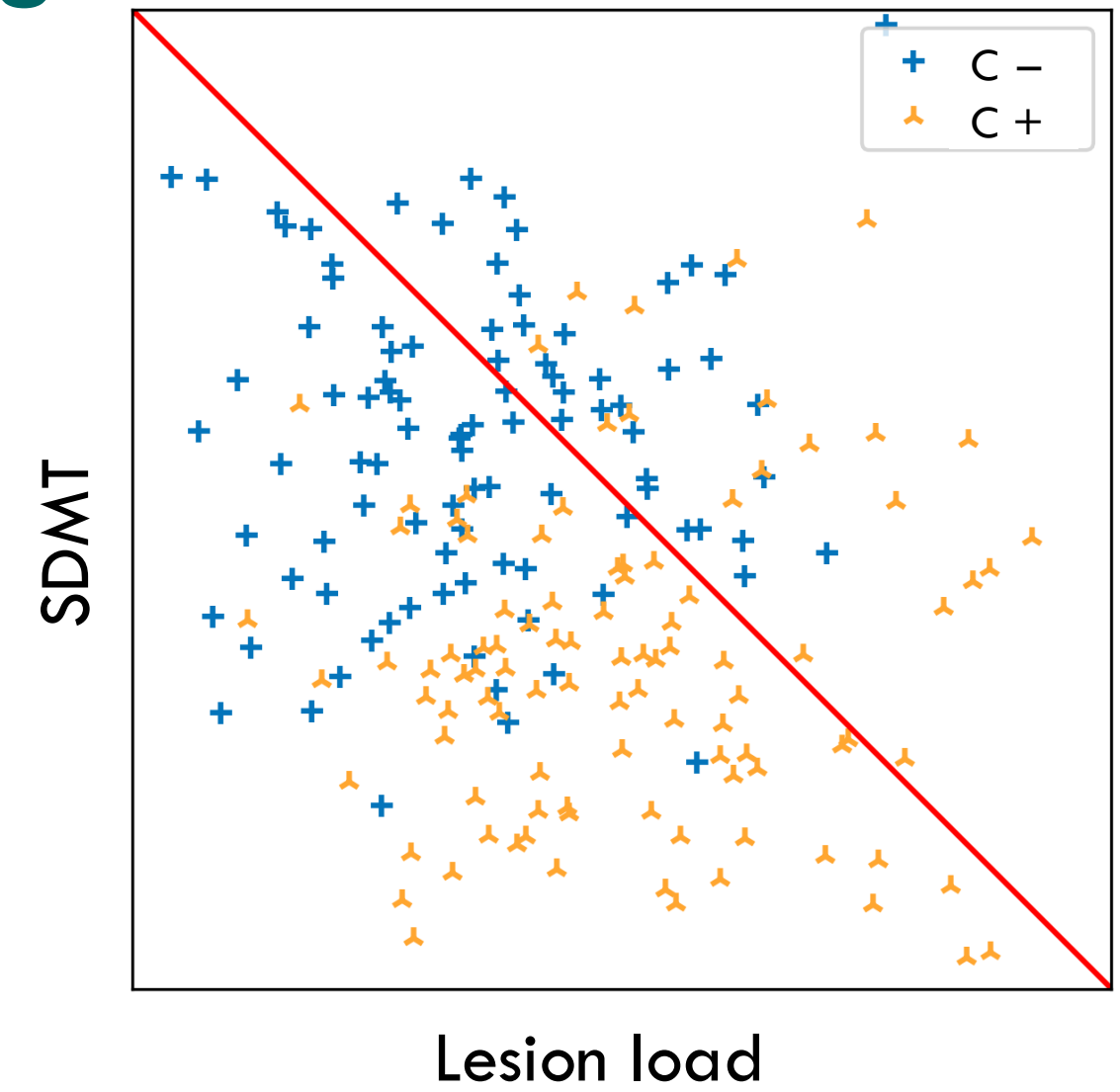
Towards a better guess

Perceptron Learning Algorithm

1. Take one misclassified point
2. Update the line
3. Repeat

Pocket Algorithm

Put best solution in your pocket



1000x updating the line

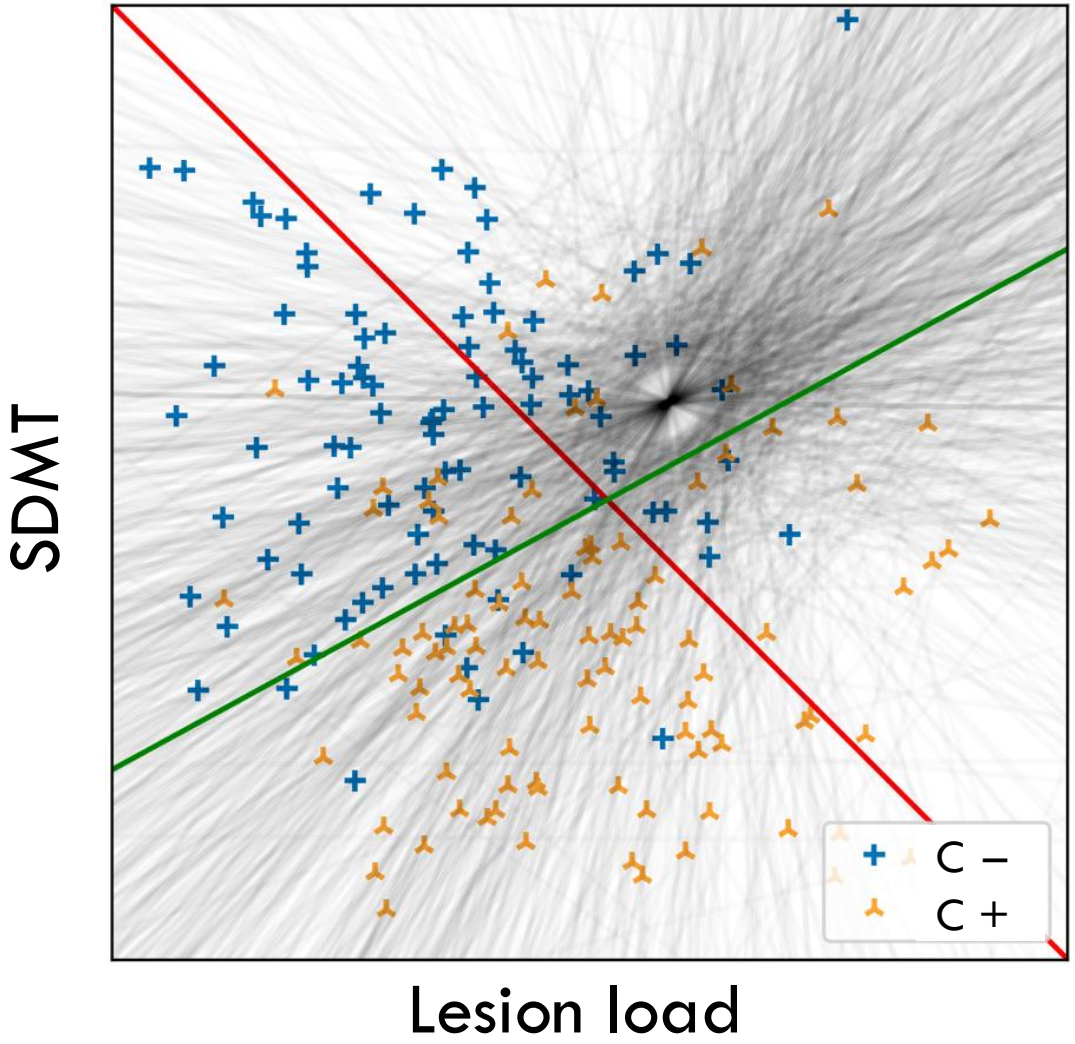
Perceptron Learning Algorithm

1. Take one misclassified point
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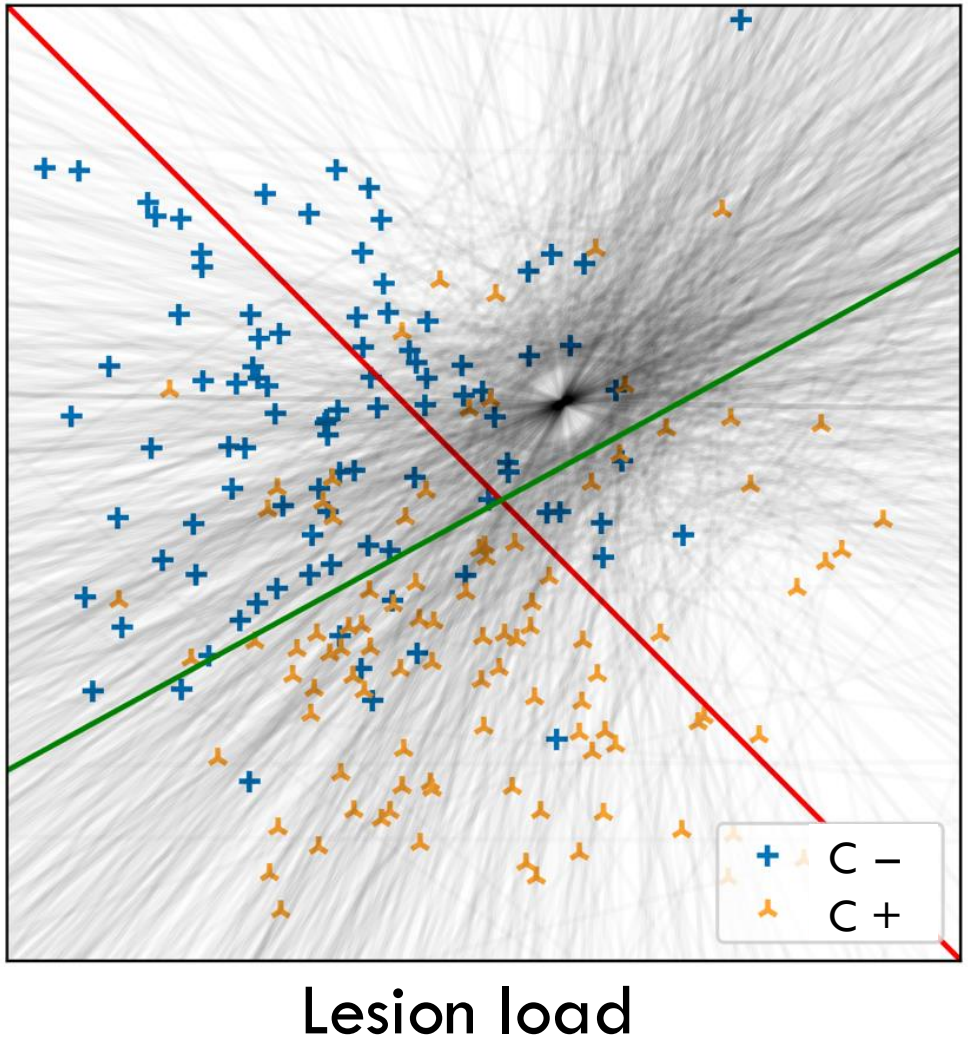
Pocket Algorithm

Put best solution in your pocket

w_S	-1.37
w_{LL}	2.5



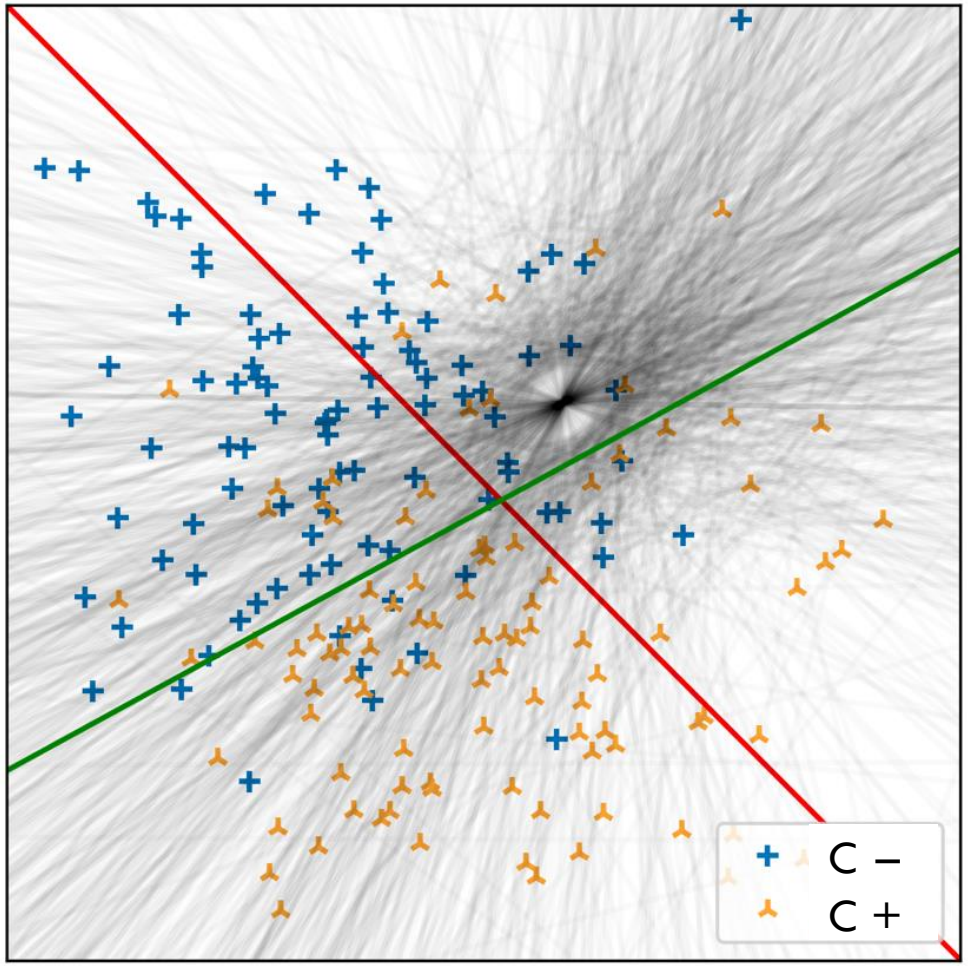
How do we know if it's the best solution?



Performance:

- Accuracy. 33 Misclassifications
- "Loss function" --> Allows training the network easier

How do we know if it's the best solution?



Performance:

- Accuracy. 33 Misclassifications
- "Loss function" --> Allows training the network easier

But it's not all performance:

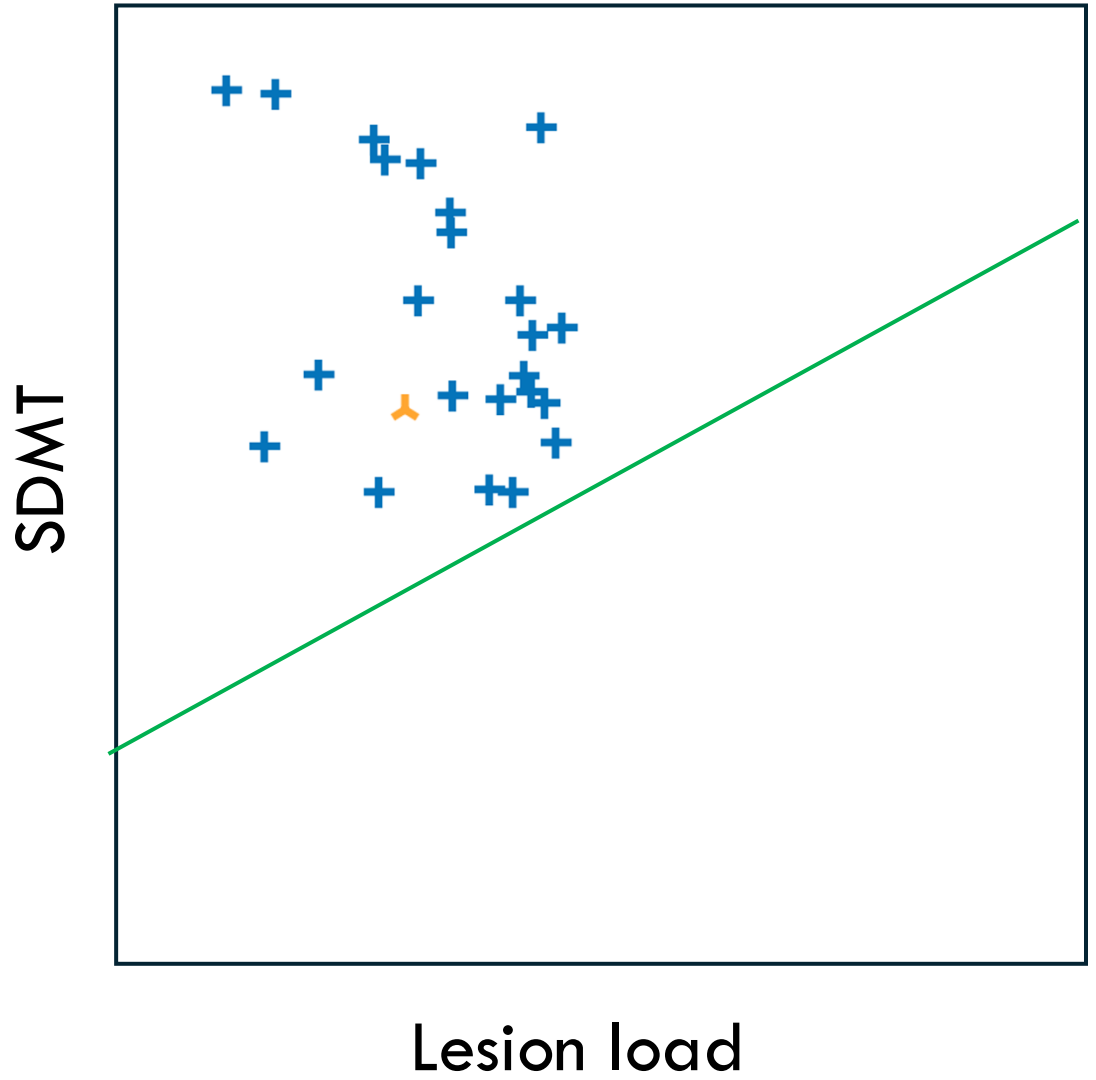
- The rise of XAI
- Lesion load twice as high impact compared to SDMT
- Does that make sense to you?
  → Trust?
- CLINICAL VALIDATION!

Another training dataset

Class Imbalance

Only 1 who converted

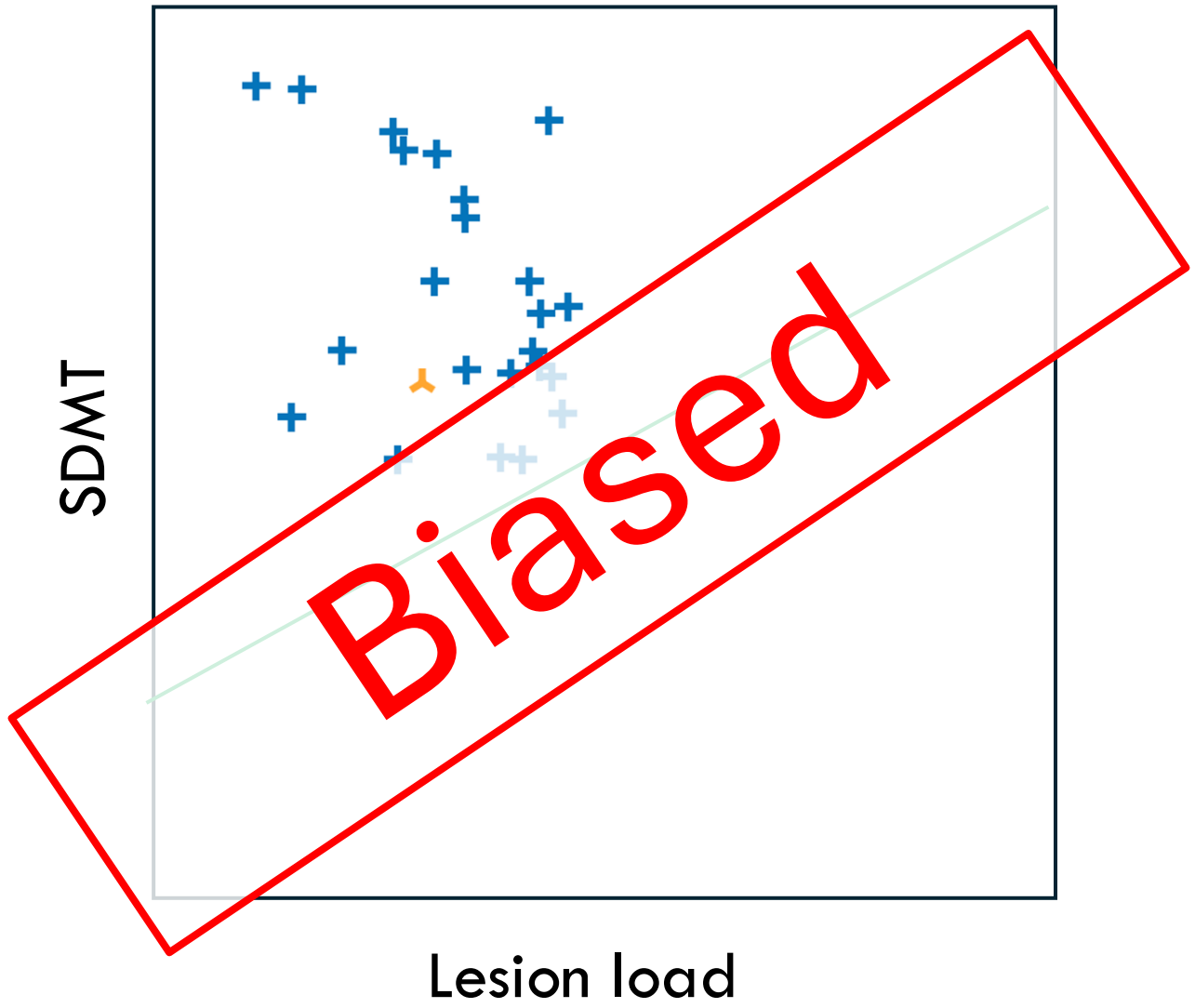
~100% accuracy!!



Another training dataset

Class Imbalance
Only 1 who converted

100% Specificity
0% Sensitivity

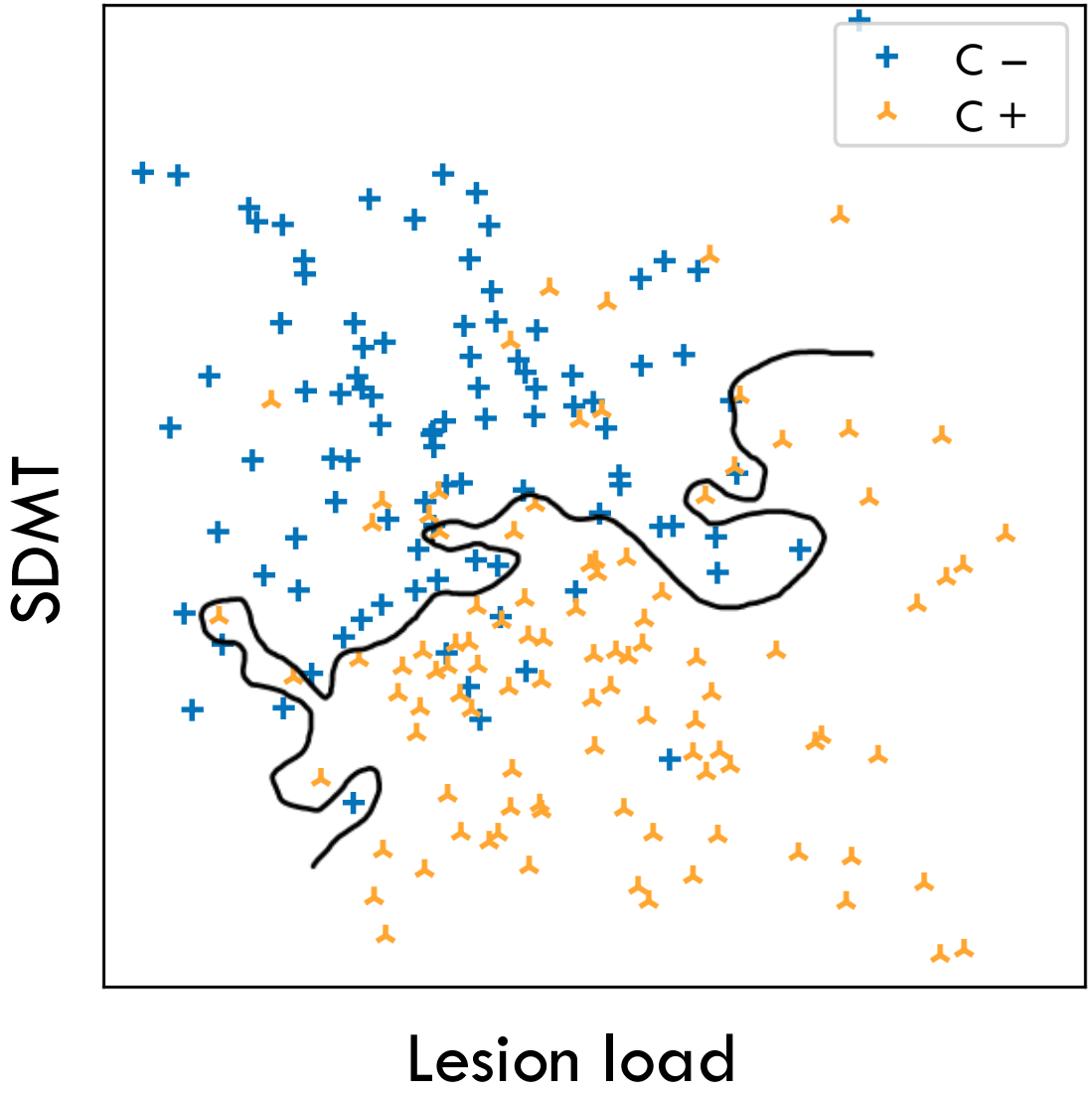


Can’t we do better?

More weights please!

Typical CNNs nowadays:

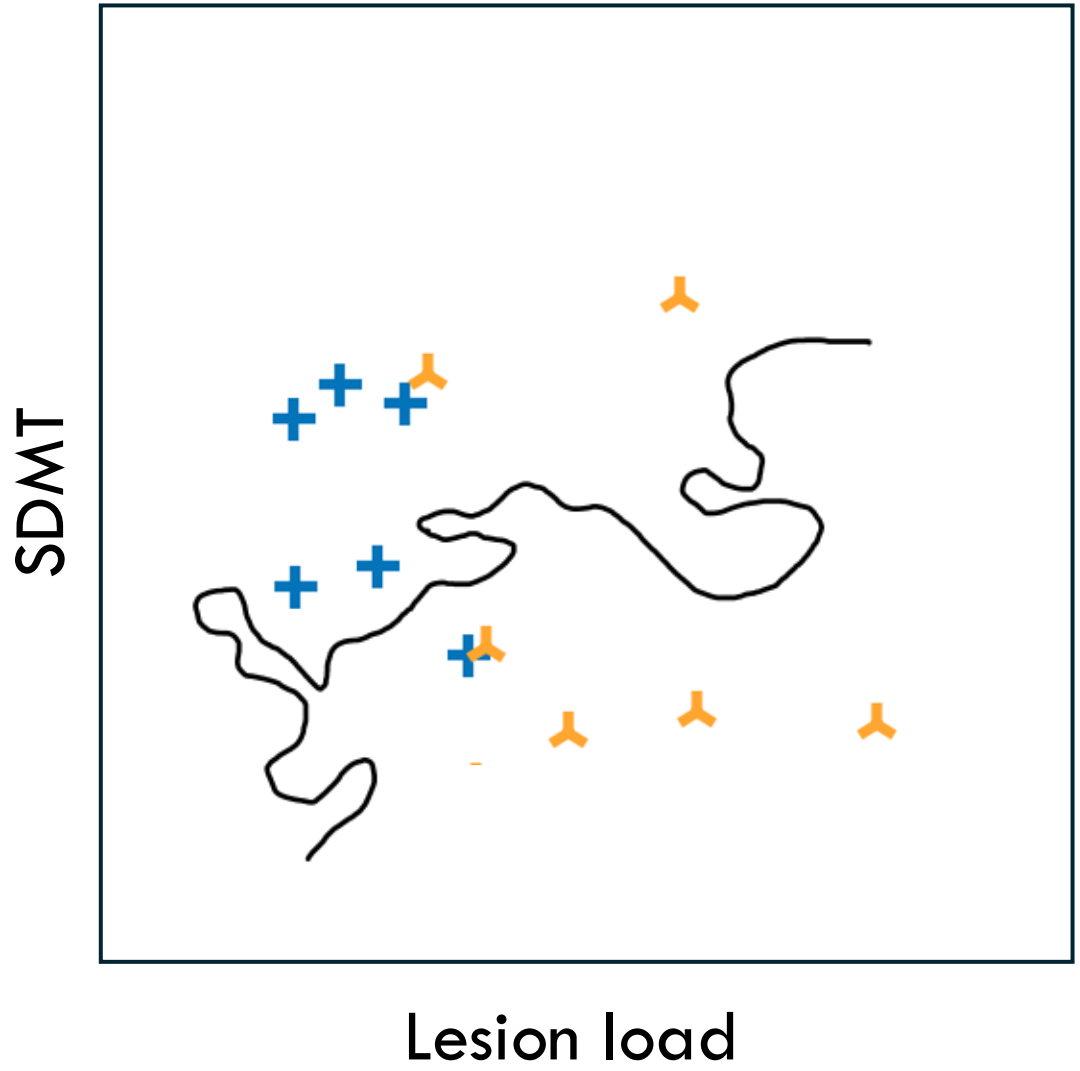
Billions



Is our model really that good?

Another clinical center

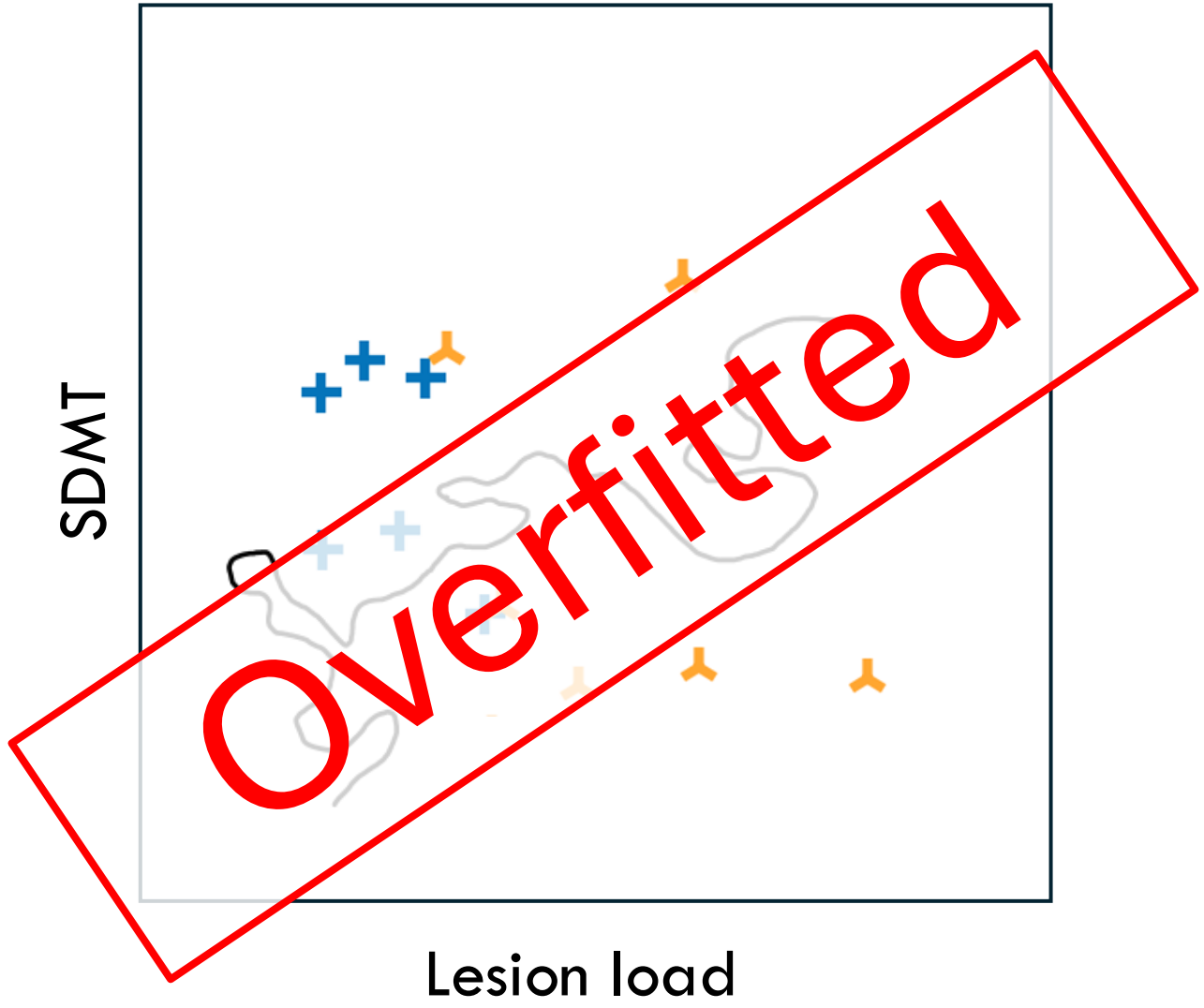
Test Dataset



Is our model really that good?

Another clinical center
Test Dataset

Avoid during training?
Validation dataset

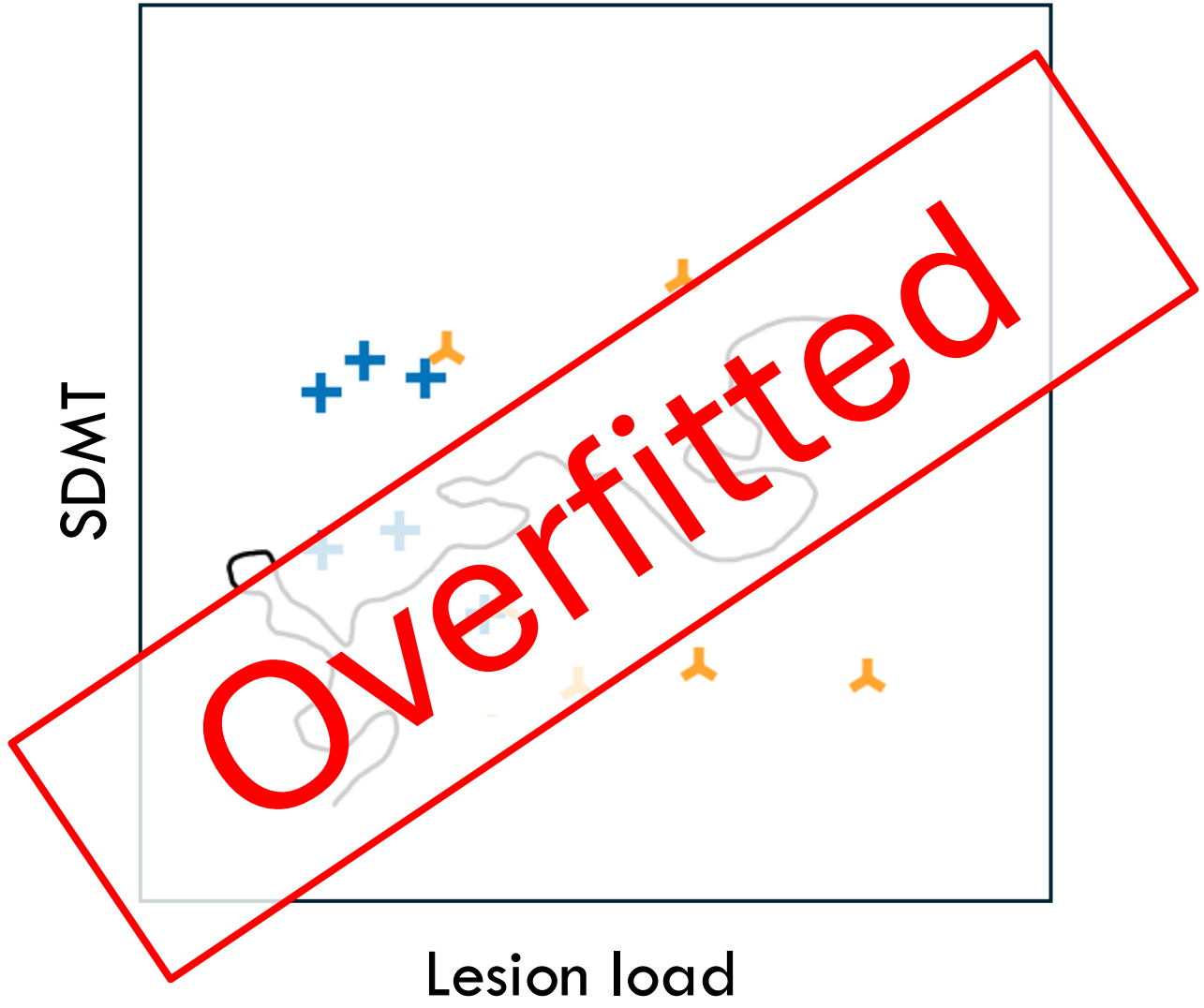


Is our model really that good?

Another clinical center
Test Dataset

Avoid during training?
Validation dataset

Concepts apply to other AI scenarios!



AI in MS

Our goals

Unite

AI x MS enthusiasts. From engineers to clinicians, statisticians to patients

Inform

About AI x MS developments, and teach bidirectionally

Co-create

Code, data and knowledge repositories driving AI for better MS care

iAIMS
international Community
for AI in Multiple Sclerosis

How to REVIEW an AI paper in MS



PlatyPeer

Inspiration:

How to read a paper involving artificial intelligence (AI)

Paul Dijkstra [1], Trisha Greenhalgh [2], Yosra Magdi Mekki,[3] Jessica Morley[4]

How to REVIEW an AI paper in MS



PlatyPeer







Screening team 👀









Writing team ✍️





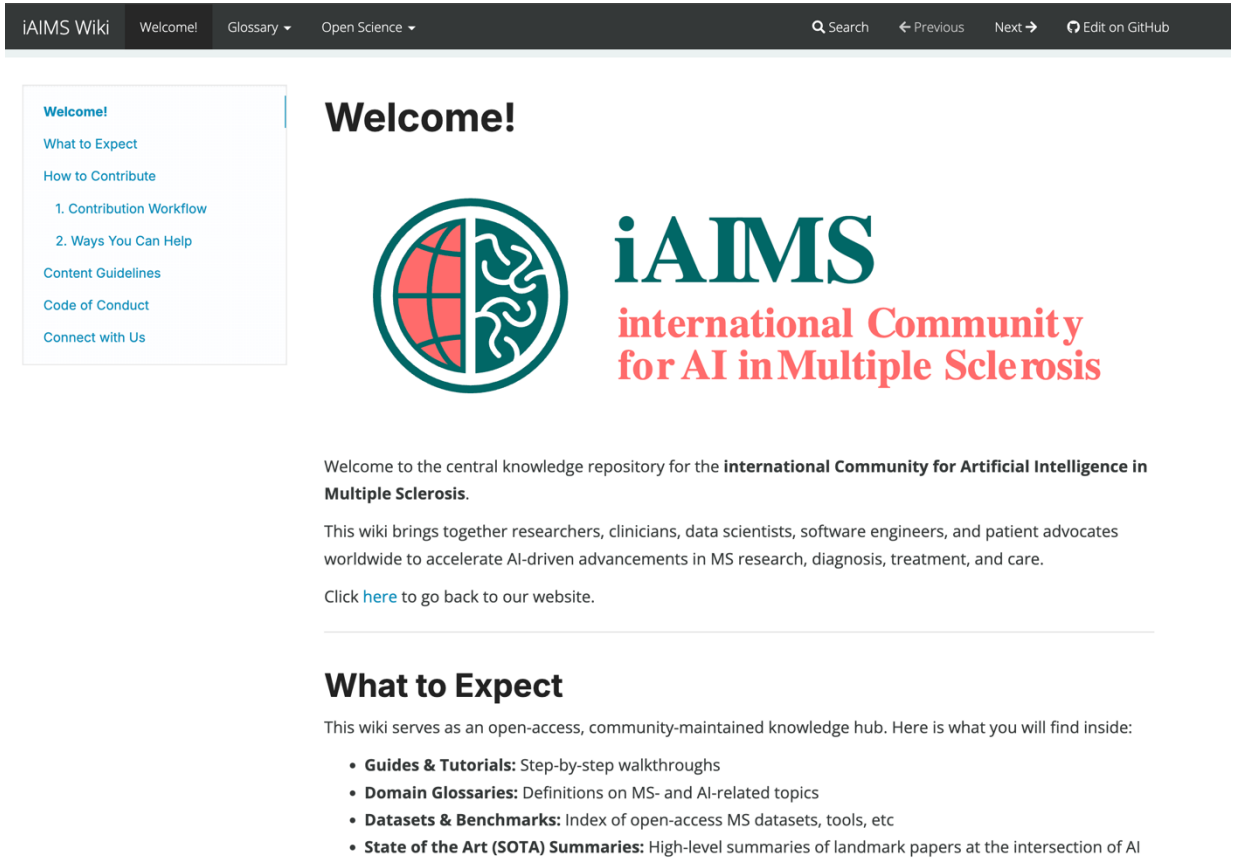
Coding team 💻

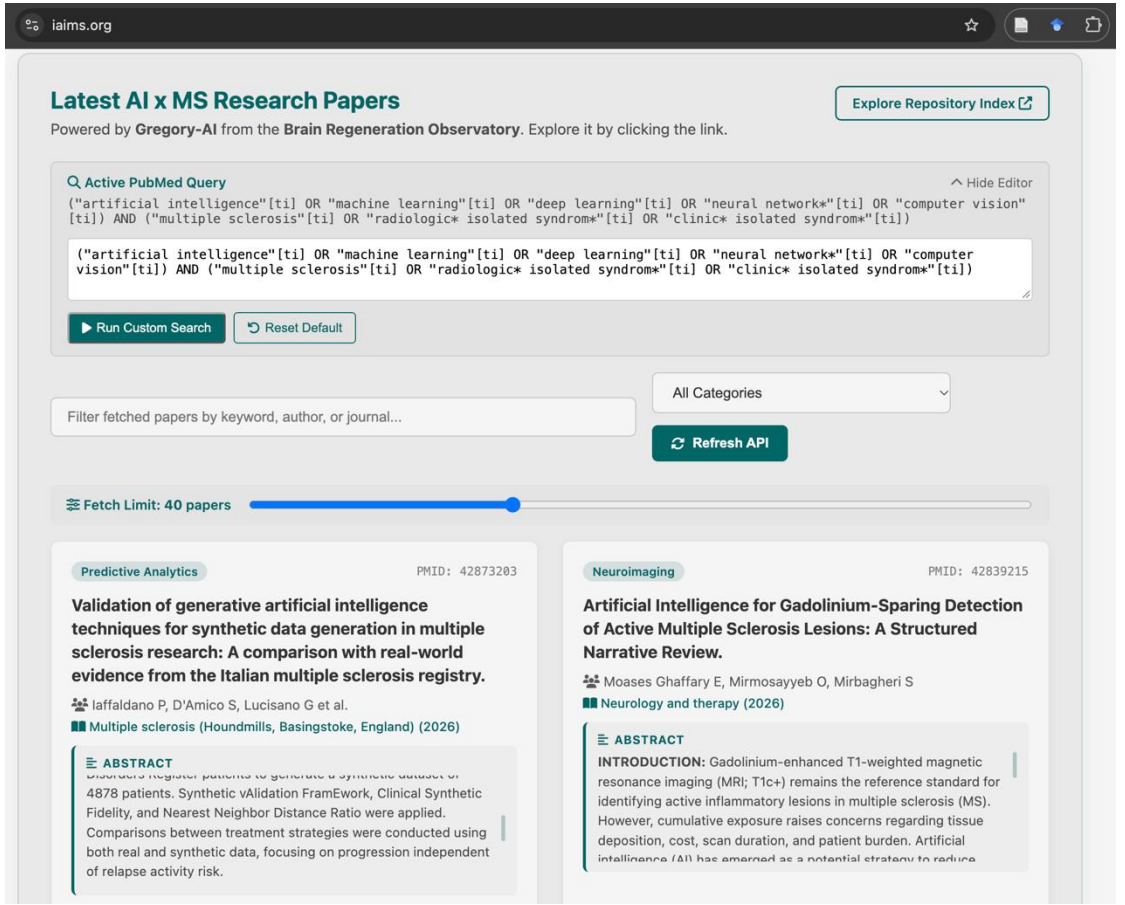
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AI literacy + latest evidence







Join iAIMS!

We need your perspective.

References and resources

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- Krizhevsky, Alex, Ilya Sutskever, and Geoffrey E. Hinton. "Imagenet classification with deep convolutional neural networks." Advances in neural information processing systems 25 (2012).
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- https://ourworldindata.org/grapher/transistors-per-microprocessor
- https://www.threads.com/@whosthat360/post/DH2ujK5y--a/sam-altman-took-to-x-formerly-twitter-saying-can-yall-please-chill-our-team-need
- https://www.nvidia.com/en-us/glossary/deep-learning/
- https://www.threads.com/@whosthat360/post/DH2ujK5y--a/sam-altman-took-to-x-formerly-twitter-saying-can-yall-please-chill-our-team-need
- https://iaims.org/platypeer/

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international Community
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Thank you

AI-SUPPORTED MODELLING
IN CLINICAL SCIENCES
RESEARCH GROUP

MSToronto2026
10th Joint ACTRIMS-ECTRIMS Meeting
OCTOBER 21-23, 2026 | TORONTO, CANADA